NCAR | HIGH ALTITUDE OBSERVATORY

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WindCube Concept of Operations

WC-SE-0003

Revision Table

Version	Editor(s)	Change Notes
1	Nir Patel	Initial Draft
2	Nir Patel	SRR Release
3	Nir Patel	Delta SRR Release
4	Nir Patel, Carsten Peterson	Updated Commissioning Activities
5	Nir Patel, Carsten Peterson	CDR Release - Added notes for shutter control - Updated solar panel diagram and added SOC diagram - Updated data budget values with prototype camera image size - Updated payload diagrams for release
6	Nir Patel	CR Release - Updated science operations diagram to indicate correct ground station location - Updated science operations to 12 months instead of 11 for stretch goal. - Added in-orbit calibration information to 3.5.
7	Nir Patel	Key Updates for RFAs 1. Removal of safe mode commanding comment 2. Adding in off-nominal operational cases to section 4 3. Discussing eclipse, impacts in section 4 4. Updating science observations to be in

		line with Qian's newest science requirements 5. Updating LEOPS and Commissioning steps and timing for first contact/separation from the deployer. Z-solar panel initial deployment changed to Z+. 6. Added etalon thermal control into systems on during telemetry mode.
8	Nir Patel, Ray Wrigley	Updated section 4 to include key eclipse operations information.

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1 - Introduction and Overview of Mission Phases

The science requirements developed during the WindCube Concept Study Report effort motivate a 12U spacecraft bus with 6U of volume allocated to the payload.

The nominal orbit for WindCube is a circular, sun-synchronous, dawn-dusk orbit at an ~500 km altitude. The science goals require the instrument aperture to be pointed along-track and cross-track in alternating orbits to measure the along-track and cross-track components of thermospheric winds.

Yaw pointing maneuvers shall be performed in the northern polar region to change orientation between orbits. WindCube will utilize Kongsberg Satellite Services (KSAT) as the commercial ground station provider.

Space Inventor, the bus provider for WindCube, will meet the mission requirement to provide the capability to downlink a minimum of 950 MB of science data per day, utilizing X-band and S-band downlink with S-band uplink. Space Inventor will provide 13300 MB nominal on-board memory for 2 weeks' worth of data in case of telemetry issues during operations.

WindCube shall have five mission phases— pre-launch, launch, commissioning, observation, and decommissioning, described in the table below.

Table 1. WindCube Mission Phases.

Phase	Duration	Description
Pre-Launch	From FRR to Launch Integration	• Transportation of WindCube and integration with deployer
Launch	From Launch Integration to Spacecraft Deployment	• Spacecraft deployment mode, detailed in the first 5-6 steps of section 2.
Commissioning	~2 weeks for S/C ~2 weeks for payload	• Establish contact and tracking • Check bus health • Power on Payload • Evaluate payload engineering data • Take test images of the 630.0nm emission and verify optimum exposure levels.
Science Observation Phase	~ 12 months (stretch goal) ~ 4 months (baseline) ~ 1 months (threshold)	• Regular uplinking schedule of scripts adjusting payload parameters as needed
Decommissioning	~1 day	• Place the satellite in deorbit orientation, detailed further in section 5.

Table 2. WindCube Mission Modes.

Operational Mode	Explanation	AOCS Config	Systems On
Deployment Mode	One time mode for satellite deployment. Includes charging as needed, detumbling, and deployment of solar panels.	Sun Pointing Mode	EPS Communications Systems AOCS
Cross-Track Observation Mode	One of WindCubes key observational orientations. Used to obtain one vector of the wind speeds. The AOCS slew maneuver shall allow WindCube to enter this orientation after downlinking.	Cross-Track Observations Mode	EPS Communication Systems AOCS Etalon Thermal Control XCAM Board Calibration Lamp Shutter driver
Along-Track Wind Observation Mode	One of WindCubes key observational orientations. Used to obtain one vector of the wind speeds. The AOCS slew maneuver shall allow WindCube to enter this orientation after downlinking.	Along-Track Observations Mode	EPS Communication Systems AOCS Etalon Thermal Control XCAM Board Calibration Lamp Shutter driver
Telemetry Mode	WindCubes communications mode. AOCS shall orient the transmitter to face the ground station while the transmitter shall downlink science data via X-band and housekeeping data via S-band. Uplinked commands will be received in this mode via S-band.	Communications Mode	EPS Communication Systems AOCS Etalon Thermal Control XCAM Board

Safe Mode	All systems except for the EPS, AOCS, and communications shall be shut off. Spacecraft shall orient in AOCS sun pointing mode.	Sun Pointing Mode	EPS Communication Systems AOCS
Deorbit Mode	AOCS shall orient Y-face nadir and the spacecraft EPS shall be disconnected.	Deorbit	AOCS EPS

Table 3. WindCube AOCS Modes.

AOCS Mode	Explanation
Sun Pointing	Maximum Power Generation Mode. AOCS shall point solar panels towards the sun while all non-essential systems shut off.
Communications Mode	AOCS shall orient for downlinking
Cross-Track Observing	AOCS shall orient for cross-track observations.
Along-Track Observing	AOCS shall orient for along-track observations.
Deorbit Mode	AOCS shall orient Y- face nadir.

2 - Payload Overview

The science payload aboard Windcube comprises a Fabry-Perot Interferometer (FPI) and an associated imaging system for capturing the transmission intensity of the etalon. Calibration images are obtained using a programmatically operated neon calibration lamp, whose photons are directed into the optical path of the FPI by an electronically operated iris with reflective shutter blades. The same shutter, when closed and with the neon calibration powered off, allows for the programmatic acquisition of dark frame images as well.

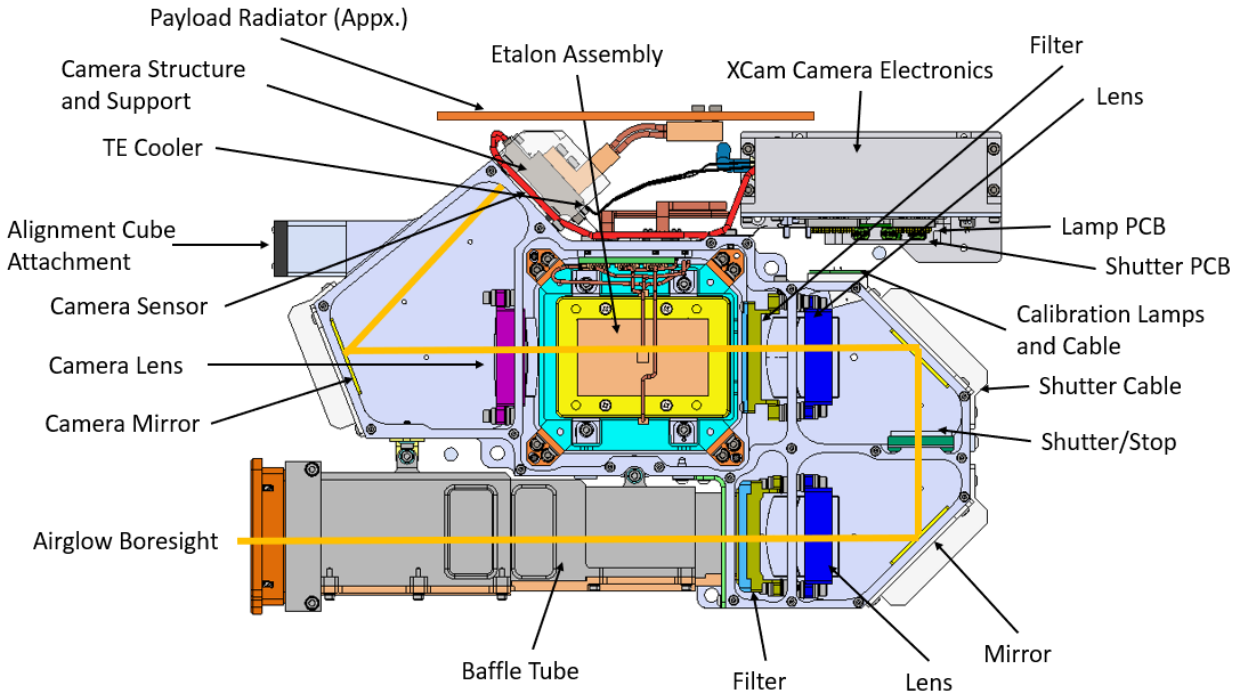


Figure 1 – WindCube Layout Diagram - Design based on 6U Payload Volume

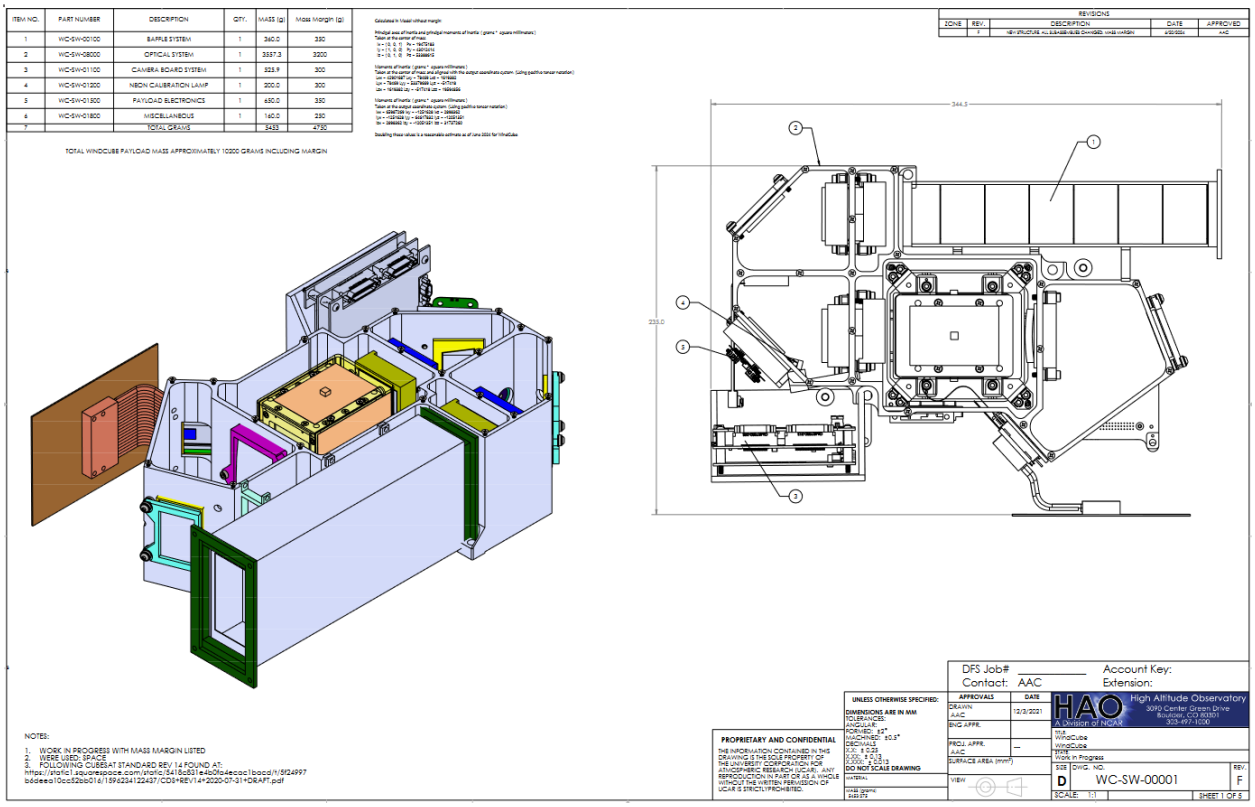


Figure 2 – WindCube Payload Diagram with Mass Breakout

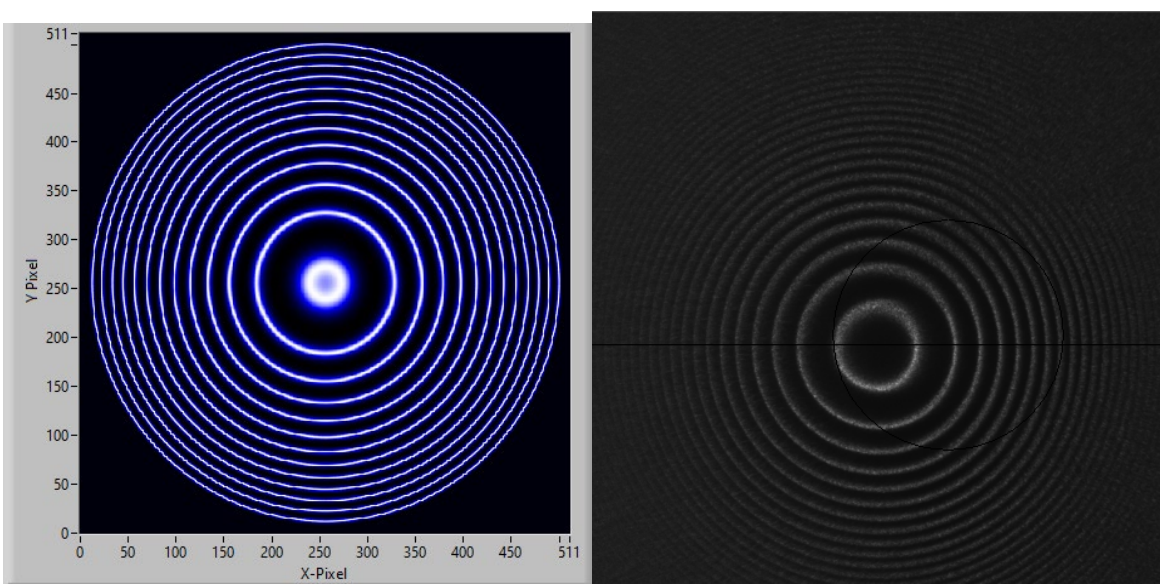


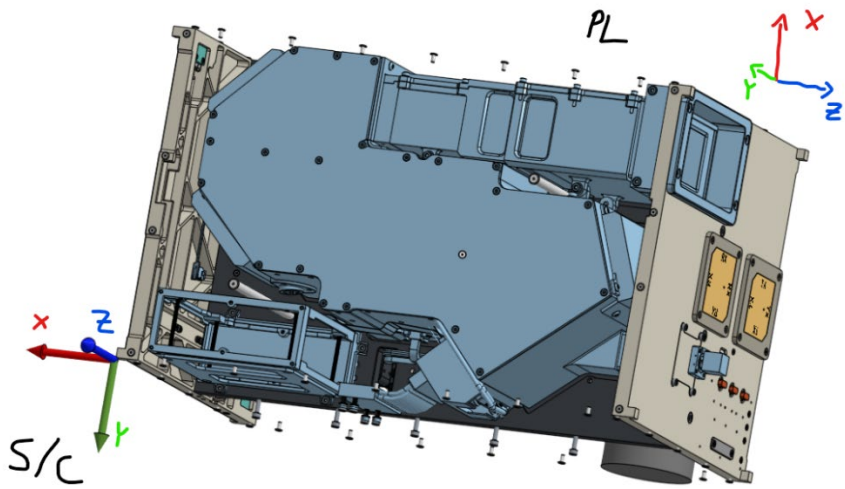
Figure 3 - (Left) A modeled FPI transmission spectrum and (Right) An actual FPI transmission spectrum using a HeNe laser. Shifts in the position of these "bullseye" patterns correspond to small doppler shifts and can be used to retrieve wind speeds from the region of the thermosphere in which the photons are generated and emitted (~200-300km)

3 – Operations During LEOP and Commissioning Phase

The Launch and Early Orbit Phase (LEOP) and commissioning phase formally starts at launcher take-off and ends when the S/C is ready for supporting the routine phase. The major activities during these phases are:

- Launch and orbit injection
- Deployment from the dispenser and automatic execution of the separation sequence
- First ground contact with spacecraft
- Early operations, i.e. spacecraft checkout and deployment of solar panels
- Commissioning, i.e. final checkout prior to entering the routine phase

Below is a diagram depicting the spacecraft coordinate axes along with the payload coordinate axes.



3.1 Launch and Orbit Injection

The spacecraft is launched to its injection-point for LEO. The spacecraft is powered-off in its dispenser during launch and both solar panels are in the stowed position. When injected, the spacecraft will be in an uncontrolled, slowly tumbling, state.

3.2 Deployment and Separation Sequence

When the launcher reaches the spacecraft orbital injection point - at a point in time controlled by the launch provider's equipment - the doors of the dispenser are opened and the spacecraft is via a spring jettisoned into space. The following separation sequence is then performed autonomously:

1. When deployed from the dispenser, the separation switches are immediately released and the Electrical Power System (EPS) with backbone power and CAN is switched on.
2. The Power Distribution Units (PDUs) will automatically power on a configured set of channels according to the *ch_on_init* parameter for each of the PDUs. The powered-on channels include the S-band radios.
3. The S-band radios used for TC/TM communication are, prior to launch, configured to listen for commands. However, the transceiver will only transmit upon request from Ground, i.e. a TC is required for the spacecraft to transmit anything. (S-Band Downlink cannot occur earlier than **18 seconds** after separation from the launch vehicle)

4. The solar panel mounted on the Z+ body panel (S/C coordinates) will be **deployed 2 min** after separation. A retry will be initiated 30 sec later, **at 2.5 min** after separation.
5. The AOCS will be set in detumble mode using only the **magnetometers 3 min** after separation. The aim is to remove any initial tumbling or spinning behavior.
6. Wait for first contact.

3.3 First Contact

Based on a TLE provided by the launch service provider, the Svalbard ground station will, at the **earliest 7 min after injection**, try to establish contact with the spacecraft using S-band. If this fails, repeated attempts to establish contact with the spacecraft will follow. After the first radio contact has been established, an updated TLE will be produced for use for the following operation of the spacecraft.

3.4 Early Operations

After ground communication has been established, initial spacecraft platform check-out will be initiated. The following activities are then performed under ground control:

1. Check-out of the Electrical Power System (EPS), making sure that the spacecraft is power positive. Verify that the solar panel mounted on the Z- body panel has been deployed and confirm solar power input.
2. Check-out the modules needed for 3-axis attitude control, excluding the Star Tracker. This may take several orbits. Upon completion, the 3-axis control will be activated and the spacecraft will be placed in Sun Pointing Mode using the momentum wheels.
3. The correct operation of the 3-axis attitude control will be verified. Again, this activity takes several orbits.
4. The solar panel mounted on the Z+ body panel will be deployed.
5. Check-out the redundant S-band radio, i.e. verify health and telemetry.
6. Check-out the housekeeping collection on the On-Board Computer.
7. Check-out the Star Tracker and enable usage of the Star Tracker in the attitude control loop.
8. Check-out of the X-band radio for downlink.
9. Pull all parameters from every module and verify with flight configuration.

At the completion of the early operations, the spacecraft is in its nominal flight configuration in terms of deployables and all platform modules have been checked-out.

3.5 Commissioning

After successful completion of the platform module check-out, the payload commissioning will take place. Payload commissioning will utilize running pre-configured test scripts which will cover the following commissioning activities.

1. Verify the thermal environment of the payload by monitoring temperature data from the payload. This activity takes several orbits, where the spacecraft is operated in both cross-track and along-track attitudes.
2. Verify the Etalon stability and adjust PID parameters for the Temperature Control Unit. Again, this is performed over several subsequent orbits to achieve a stable temperature at the Etalon.
3. Check-out the XCAM camera. Turn the camera on/off, power TEC board on & verifying what level of cooling the focal plane cooling system can achieve. Configure camera (binning, exposure time, gain adjustment, TEC set point) and obtain an exposure.

4. Check-out the Calibration Lamp System. Verify ability to select boards and power lamps and verify housekeeping data.
5. Check-out the Shutter System. Verify ability to open, close, and enable the shutter. Housekeeping data verification.
6. Combined payload system commissioning. Close the shutter and acquire a dark image. Turn on the lamps and acquire a calibration image. Turn off the lamp, open the shutter, acquire an airglow image. Downlink associated housekeeping data.

In-orbit calibration of the spacecraft star tracker and instrument detector will be performed as described below:

1. *The spacecraft attitude is adjusted such that the center of the XCAM imager FoV is oriented towards a bright planet (e.g., Jupiter), while the star tracker is oriented away from earth.*
    1. *Note that this will require viewing a star that does not risk exposing the image detector towards the Sun.*
2. *XCAM and star tracker data is downlinked.*
3. *Alignment between the star camera and the XCAM payload is estimated based on the data.*
4. *New alignment data is added to the pre-launch alignment of the AOCS SW, and the new values are uplinked.*
5. *Repeat steps 1-5 until the alignment is within the required tolerance.*

After this point. Commissioning will be complete, and the spacecraft will begin its Science Observation Phase.

4 - Detailed Operations during Science Observations Phase

WindCube has been designed with a nominal terminator orbit with a Local Time of Ascending Node (LTAN) of 0600 +/- 1 hour at an altitude from 500 km to 600 km at launch. WindCube selected this nominal orbit to provide a consistent thermal environment for the instrument system. Per analysis of the orbit, over a 13-month mission WindCube will experience an eclipse season during northern winter. The duration of this season is approximately 22 minutes for an LTAN of 0600, increasing to approximately 26 minutes for LTAN values +/- 1 hour from 0600. Below are two sequences of spacecraft and payload activities corresponding to **nominal** operations outside of eclipse season and **reduced** operations when at least part of the orbit is in eclipse.

4.1 - Orbital Events for Nominal Operations

The following represents a **nominal** set of spacecraft and payload commanded activities post commissioning phase outside of eclipse season.

1. Control spacecraft attitude to position boresight along-track – 180 seconds
2. Shutter system activates to close the shutter – 11 milliseconds
3. Acquire ~1-5 dark images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
4. Neon lamp system activates illumination– 1 second
5. Acquire ~1-5 calibration images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
6. Neon lamp system deactivates – instantaneous
7. Shutter system opens the shutter – 11 milliseconds
8. Acquire ~448 science images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 4,480 seconds
    a. Note that the number of science images will be adjusted based on the eclipse environment and/or presence of geomagnetic storm while meeting the baseline observational latitude requirements.
    b. During this transfer to the telemetry queue, metadata will be added to science images containing timestamps, spacecraft attitude and position, and camera configuration data.
9. Shutter system closes the shutter – 11 milliseconds
10. WindCube will then continue to orbit until receiving command from the KSAT's ground station in Svalbard for data downlink
11. Perform ~20-degree pitch maneuver to position transmitting antenna directly towards nadir – 180 seconds
12. Transmit all science and engineering data residing in the telemetry queue, receive any uploaded scripts/commands from the ground station – 480 seconds
    a. Note that this step will occur up to 5x per day
13. Perform 90 Yaw maneuver (and reverse 20 pitch maneuver) to position boresight across-track – 180 seconds
14. Repeat Steps 2-12 accounting for any additional downlink passes with along-track and cross-track orientations switching per orbit.

4.2 - Orbital Events for Reduced Eclipse Operations

The primary impact of an eclipse is a reduced power availability due to the inability to charge batteries in eclipse. Mission operations will be required to incrementally reduce the duration of science observations as eclipse duration increases. The time that becomes available will then be used to charge the batteries.

With a morning LTAN, eclipses always take place over the northern hemisphere. Data downlink, which happens on average slightly less than one every three orbits, will generally be scheduled to take place during eclipse. Science observation placement will be biased toward the equator where the highest value data is obtained to support Windcube Science Goals 1 and 2, though it is possible to include polar observations that support Science Goal 1 even during eclipse season. During periods outside of eclipse season, observations over the south pole are reincorporated to enhance support for Science Goal 1.

Figure 4 illustrates the eclipse duration for a mission where the science phase begins in July 2026. This will be updated once WindCube has a confirmed launch window.

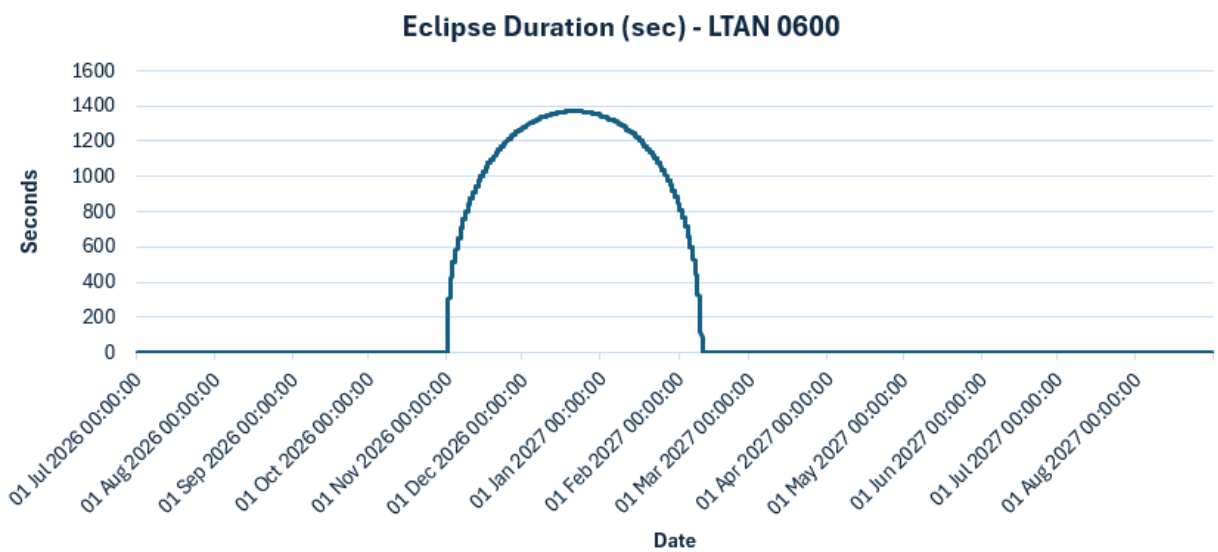


Figure 4 - Approximate eclipse durations over a 13-month mission beginning in Summer

The following represents a **reduced** set of spacecraft and payload commanded activities post commissioning phase during eclipse season. An eclipse duration of 22 minutes (the maximum for LTAN 0600) is chosen for illustration. This description begins as the observatory enters the eclipse while proceeding north toward the polar region. This narrative covers two 94-minute orbits. Please note the following is just an example of what operations may look like depending on the eclipse timing.

1. Observatory enters eclipse.
2. Perform ~20-degree pitch maneuver to position transmitting antenna directly towards nadir – 180 seconds
3. Transmit all science and engineering data residing in the telemetry queue, receive any uploaded scripts/commands from the ground station – 480 seconds
    a. Note that this step will occur only 5x per day

4. Control spacecraft attitude to position boresight along-track – 180 seconds
5. Shutter system activates to close the shutter – 10 milliseconds
6. Acquire ~1-5 dark images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
7. Neon lamp system activates to illuminate one of three neon lamps – 1 second
8. Acquire ~1-5 calibration images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
9. Neon lamp system deactivates – instantaneous
10. Shutter system opens the shutter – 10 milliseconds
11. Acquire ~150 science images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – ~1500 seconds
    a. Note that the number of science images will be adjusted based on the eclipse duration.
    b. The observatory exits eclipse roughly halfway through this observing period.
    c. During this transfer to the telemetry queue, metadata will be added to science images containing timestamps, spacecraft attitude and position, and camera configuration data.
12. Reorient observatory to direct solar arrays at Sun.
13. Maintain charging attitude - 32 minutes
14. Control spacecraft attitude to position boresight along-track – 180 seconds
15. Shutter system activates to close the shutter – 10 milliseconds
16. Acquire ~1-5 dark images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
17. Neon lamp system activates to illuminate one of three neon lamps – 1 second
18. Acquire ~1-5 calibration images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – 10-50 seconds
19. Neon lamp system deactivates – instantaneous
20. Shutter system opens the shutter – 10 milliseconds
21. Acquire ~132 science images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – ~1320 seconds
22. **At this point, the observatory has completed one orbit and re-enters eclipse.**
23. Optional (depends on the desire to acquire cross-track data on every alternate orbit) - Perform 90 Yaw maneuver to position boresight across-track – 180 seconds.
24. Acquire ~204 science images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – ~2040 seconds
    a. The observatory exits eclipse during this observation.
25. Reorient observatory to direct solar arrays at Sun.
26. Maintain charging attitude - 30 minutes
27. Control spacecraft attitude to position boresight along-track – 180 seconds
28. Acquire ~150 science images, transfer to telemetry storage buffer queue on the spacecraft flight computer system – ~1500 seconds
29. **At this point, the observatory has completed a second orbit and re-enters eclipse.**

To support the steps above, the XCAM camera system will include the following script for the shutter system:

Table 4. Shutter Scripts

Script 1:	20 millisecond long pulse send over the open shutter GPIO line
Script 2:	20 millisecond long pulse send over the close shutter GPIO line
Script 3:	1. XCAM will flush the camera (transfer the image) 2. Start a new exposure 3. Wait 50 milliseconds 4. Open the shutter by sending a high pulse on the open shutter GPIO line – 20 millisecond long pulse 5. Wait for the exposure duration (whatever length has been commanded for this) 6. Close shutter by sending a high pulse on the close shutter GPIO line – 20 millisecond long pulse 7. Wait 50 milliseconds after that pulse 8. Transfer image
Script 4:	1. XCAM will flush the camera (transfer the image) 2. Wait for the exposure duration (whatever has been commanded for this) 3. Transfer image

4.3 - Science Data Collection Summary

WindCube will collect wind data (Fabry Perot etalon transmission images) in an along-track orientation for one orbit. Upon reaching a ground station contact position (e.g., Svalbard), WindCube will enter the telemetry mode. During telemetry mode, a schedule of commands may be uplinked, replacing pre-configured schedules as needed. In addition, engineering data and Level-0 science data products (defined within RTM) will be downlinked while in telemetry mode.

Upon completing a ground station pass and exiting telemetry mode, WindCube will transition into the cross-track orientation for data collection during the next orbit. WindCube will continue to rotate through the along-track, telemetry, and cross-track modes for the duration of the observation schedule, unless interrupted by time for housekeeping activities or anomalous conditions which will then prompt WindCube to enter safe mode.

Preconfigured and uploaded command schedules may also include setting WindCube into its calibration mode. Payload calibrations consist of acquiring both a dark image (by closing a shutter) as well as a calibration image where the Fabry Perot etalon is illuminated with an onboard neon emission lamp source.

Should WindCube enter a severe fault condition, a safe mode will be entered which prioritizes power saving and ground contact until the condition can be recovered. This safe mode will also function as the sun-pointing mode for the purpose of allowing WindCube to charge while conducting issue resolution. WindCube shall also have flexibility to include up to 4 dedicated charging orbits per day if required for operations.

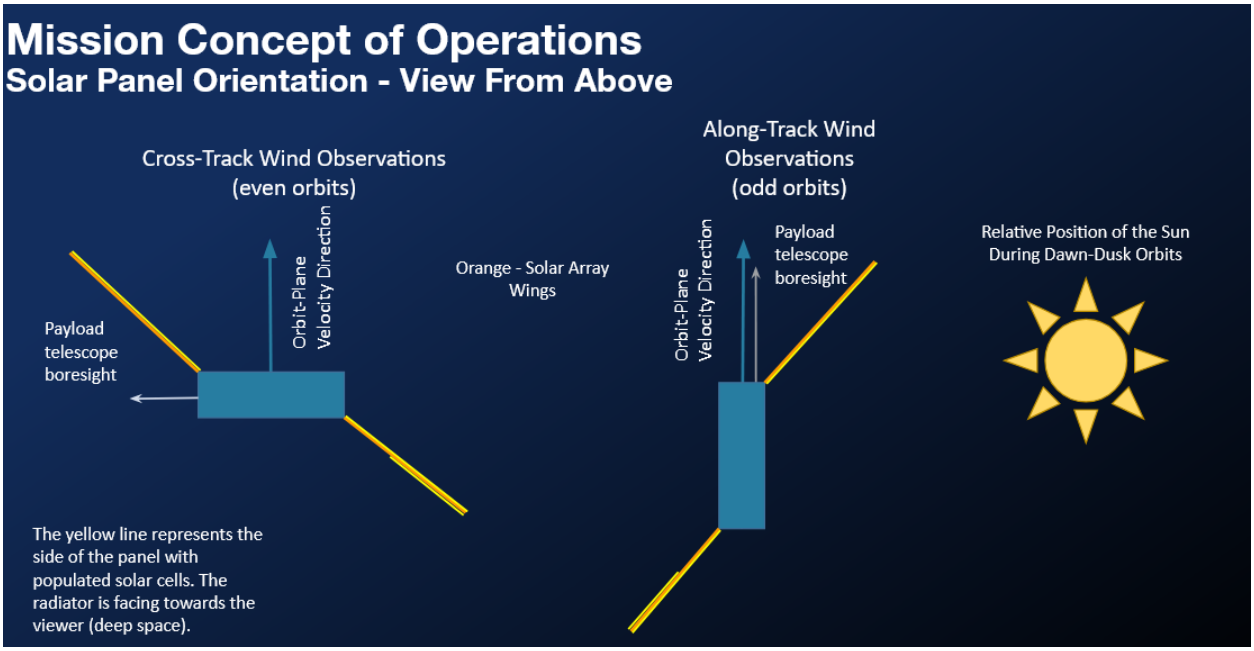


Figure 4 - Depiction of the two planned orientations of the WindCube spacecraft on alternating orbits. The two orientations are required to measure thermospheric wind speeds in and across the direction of spacecraft travel. ***The solar panels shall be mounted so that when deployed, achieve the maximum power generation when averaged over two orbits.***

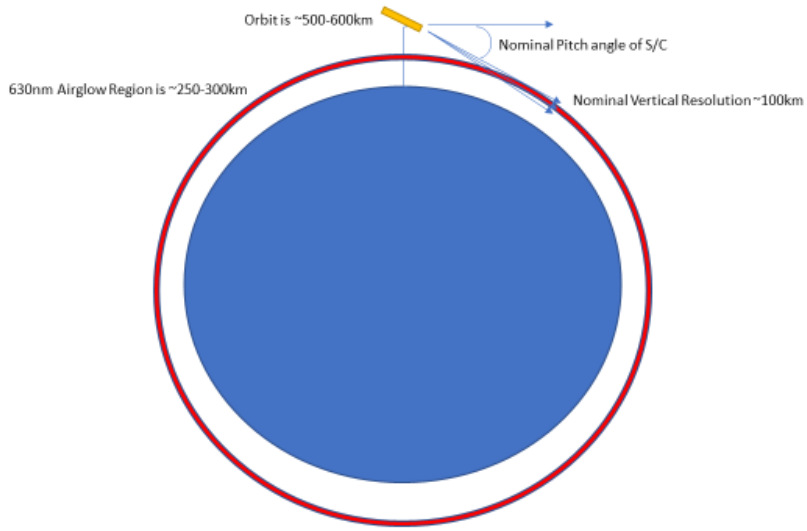
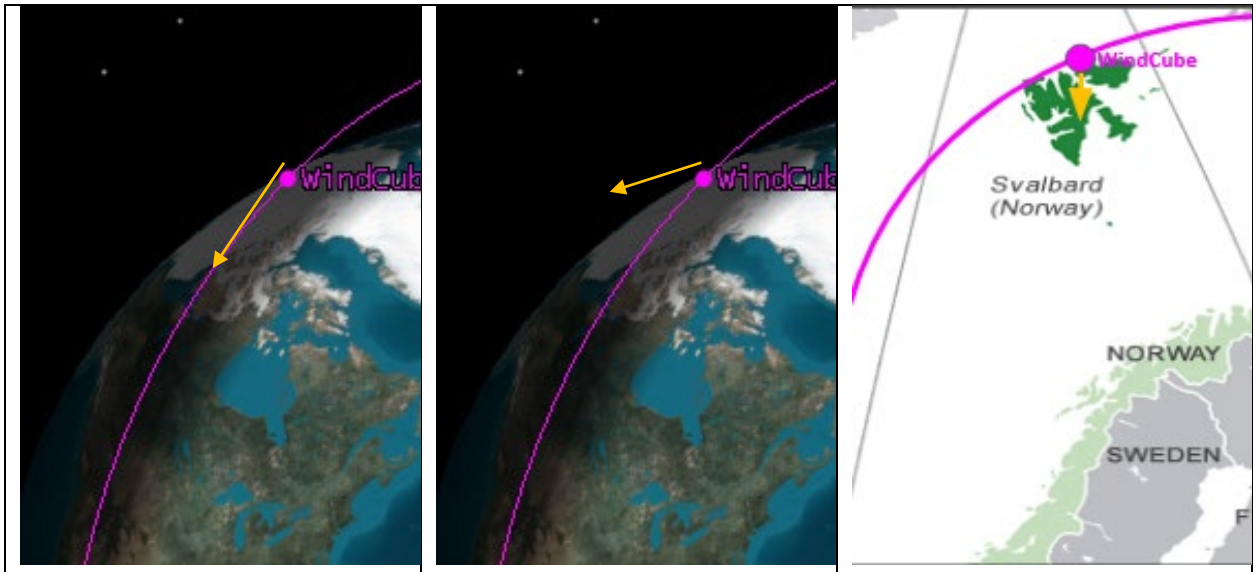


Figure 5 - Depiction of the spacecraft pitch required to sample the emitting region of the thermosphere between 250-300 km.



Along-Track Science Data Collection	Cross-Track Science Data Collection	Data Downlink Pass and Payload Calibration Procedures

Figure 6 - Depiction of the three basic attitude positions required during the WindCube science operations phase.

5 - Mission Operations

WindCube Mission Operations shall be operated through the WindCube Mission Operations Center (MOC) by Space Inventor. The MOC shall serve as the communication interface between the ground station(s) and HAO's Science Operations Center (SOC). Operators for the MOC shall come from Space Inventor, while operators for the SOC shall be the project PI/Co-I and the project software engineering lead from HAO. The nominal flow of data is depicted below in Figure 8.

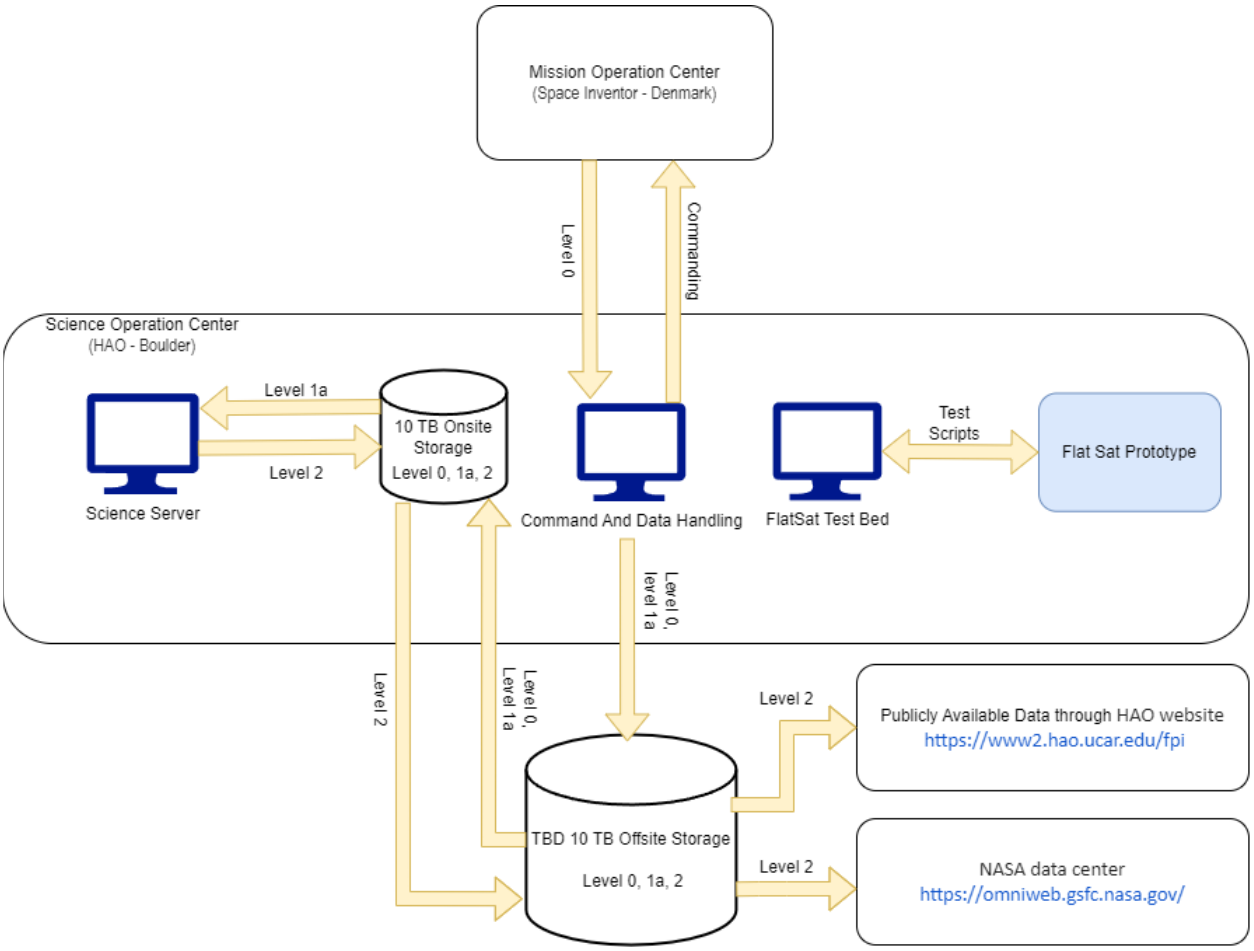


Figure 7 – Science Operation Center Topology

Once WindCube is in operation, following commissioning, the sequence of events below shall represent the operating procedure between the ground station(s), the MOC, and the SOC:

1. Once per week, the SOC shall produce a science observation script with the following parameters and send this script to the MOC.
    a. Change camera system parameters. E.G. Exposure length, TEC temperature, binning.
    b. Change frequency of dark and calibration images.
    c. Changing parameters for the etalon temperature control system.
    d. Power cycling camera
    e. Selecting calibration lamp for utilization

2. The MOC shall take the received script and prepare it for upload to WindCube through the ground station.
3. The ground station shall receive downlinked data from WindCube, including the images and spacecraft/payload housekeeping data.
4. The ground station shall transfer this downlinked data to the MOC, which will then utilize necessary telemetry data to ensure WindCube's functionality.
5. The MOC shall transfer the image and payload housekeeping data to the SOC.
6. The SOC will receive the image and payload housekeeping data and perform data analysis to generate level 1 and level 2 WindCube global wind speed maps from the level 0 image data.
7. All data generated by the SOC shall be transferred to a database.
8. Repeat steps 1-7.

6 - Decommissioning Phase

FCC requirements dictate that satellite missions be deorbited within 25 years of termination of the primary mission. WindCube's estimated orbital lifetime from the NASA Debris Analysis Software (DAS) is 4.641 years (WindCube CR ODAR) and is compliant with the requirement to reenter within 25 years after the mission completion or 30 years after launch.

Taking into consideration the adjusted rules imposed by the FCC after the funding of the WindCube project, WindCube plans to orient with the -Y face of the CubeSat prograde in order to maximize surface area exposed to drag in order to reduce the orbital lifetime after the mission lifetime has come to an end.